

MUSTANSIR MOHAMMADI

BCA Student | Software Development | Game AI Engineering

Python • C++ • Django • APIs • SQL • Data Structures & Algorithms

Phone: 9636929753 | Email: mum2524086@sicsr.ac.in | LinkedIn: Mustansir Mohammadi |

GitHub: github.com/MUSTANSIRBM/python-portfolio

PROFILE

BCA student specializing in **Game AI Engineering and systems programming**, with hands-on implementation experience in C++. Developed knowledge across *A*, *Dijkstra*, *NavMesh*, *Funnel Algorithm*, *Behavior Trees*, *FSMs*, *steering behaviors*, *flocking*, *Blackboard architecture*, and *memory management**. Complementing this with Python and backend development experience, I enjoy building complete systems from the underlying data structures and algorithms upward.

TECHNICAL SKILLS

Languages	Python, C++, SQL
Backend	Django, REST APIs, JSON, backend development fundamentals
Core CS	OOP, Data Structures, Algorithms, memory management, file I/O, ETL, data persistence
Game AI	A*, Dijkstra, BFS/DFS, NavMesh, Funnel Algorithm, path smoothing, FSM, Behavior Trees, Steering, Flocking, Blackboard
Tools & Practices	Git / Version Control, os, shutil, CLI application development, debugging and validation

PROJECTS

Complete Game AI Engineering System — C++ | Jun 2026 – Present

- Designed an implementation-focused Game AI portfolio spanning 48 topics across data structures, pathfinding, 3D navigation, decision systems, movement, and AI memory.
- Implemented/covered A* pathfinding, Dijkstra, NavMesh navigation, Funnel path smoothing, FSMs, Behavior Trees, steering/flocking, and Blackboard communication.
- Applied C++ memory-management concepts including Rule of 5, RAII, deep copy, move semantics, and type-safe Blackboard storage.

Python Software Engineering Portfolio — 16 CLI Applications | 2025

- Built 16 functional command-line applications demonstrating OOP, data persistence, ETL-style load/save logic, JSON processing, file I/O, and business logic.
- Developed projects including University Course Management, Train Booking, Project Management, Student Management, ATM, Task Manager, Quiz Engine, and My_Sorter.
- Used custom decorators, inheritance, nested data structures, os/shutil file operations, and JSON serialization.

Backend Development Learning Projects | 2025 – 2026

- Applied concepts from Meta-authorized Coursera coursework covering backend development, Django, APIs, databases, Python, and version control.
- Developed a foundation for building structured backend applications using Python, Django, APIs, JSON, SQL/database concepts, and Git.

EDUCATION

Course / Specialization	Institute / College	University / Board	Year	CGPA / %
BCA	Symbiosis Institute of Computer Studies and Research	BCA	2025 – Present	Sem 1: 8.5 Sem 2: 8.43
Class XII	New look Institute	CBSE	2024	69.8%
Class X	MSB(Educational Institution)	ICSE	2023	73.7%

CERTIFICATIONS

- APIs** — Meta / Coursera | 23 Jan 2026
- Django Web Framework** — Meta / Coursera | 10 Jan 2026
- Introduction to Databases for Back-End Development** — Meta / Coursera | 30 Nov 2025
- Version Control** — Meta / Coursera | 20 Nov 2025

- **Programming in Python** — Meta / Coursera | 17 Nov 2025
- **Introduction to Back-End Development** — Meta / Coursera | 14 Nov 2025

ACHIEVEMENTS & PORTFOLIO HIGHLIGHTS

- Completed a 48-topic Game AI engineering portfolio with implementation-level coverage across six major domains.
- Built a 16-application Python software engineering portfolio demonstrating OOP, ETL/data persistence, decorators, JSON, and file-system automation.
- Game AI portfolio includes advanced topics such as Funnel Algorithm (10/10), Behavior Trees (9/10), and NavMesh (9/10).

ADDITIONAL INFORMATION

Languages	English,Hindi
Hobbies	Hobbies & Interests Creative Writing Programming & Algorithmic Problem Solving Interests Building experimental software projects
Date of Birth	12/03/2006